Trajectory Tracking while Stabilizing an Inverted Pendulum on a Quadcopter Using Adaptive Model-Predictive Control

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***Abstract* — We combine the canonical problem of stabilizing an inverted pendulum with quadcopter trajectory tracking to study the control of a top mounted payload in an UAV, referred to as the *flying inverted pendulum*. Traditional nonlinear model predictive controllers (MPC) have proved effective for both agile quadcopter maneuvers as well as inverted pendulum stabilization. However, due to its model based nature, the feedback system suffers significant less in performance under modelling uncertainties and external disturbances. In this paper, we present a novel strategy to control this UAV model by utilizing an *adaptive model-predictive controller*. A nominal nonlinear MPC is augmented with an $\mathcal{L}_1$ adaptive controller which performs online estimation of the model uncertainties, especially those unknowns pertaining to the pendulum configuration. The adaptive controller continually compensates for modelling errors, resulting in considerable improvements compared to the uncertainty-sensitive baseline controller. Experiments are conducted in simulation to validate the performance of the control laws in practice compared to existing control techniques.**

***Index Terms* — Flying inverted pendulum, model predictive control, adaptive control, quadcopters.**

I. Nomenclature

$\mathcal{I}$	=	interial/ world frame
$\mathcal{B}$	=	body frame of the quadcopter
$\mathcal{C}$	=	frame attached to the pendulum
$\boldsymbol{e}_i$	=	unit vectors of frame $\mathcal{I}$
$\boldsymbol{b}_i$	=	unit vectors of frame $\mathcal{B}$
$\boldsymbol{c}_i$	=	unit vectors of frame $\mathcal{C}$
$\boldsymbol{O}$	=	origin of the inertial frame
$\boldsymbol{G}$	=	center of mass of the quad
$\boldsymbol{P}$	=	point representing the payload
$\boldsymbol{r}_{a/b}$	=	distance vector with root at b and tip at a (b to a)
${}^{\mathcal{X}}\boldsymbol{v}_{a/b}$	=	velocity vector for distance b to a when observed from some frame $\mathcal{X}$
$\boldsymbol{R}_{\mathcal{X}}^{\mathcal{Y}}$	=	rotation matrix from some frame $\mathcal{X}$ to another frame $\mathcal{Y}$
g	=	gravity
m_G and m_P	=	mass of the quadcopter and the payload, respectively
d and l	=	length of each of the four arms of the quadcopter and the cable, respectively
$\mathbb{I}_{\mathcal{B}}$	=	principal inertia tensor of the quad in $\mathcal{B}$ with diagnoal entries $I_{1\dots3}$
C_L	=	lift coefficient of propellers
C_D	=	drag coefficient of propellers
q	=	generalized coordinates
Q_j	=	generalized force corresponding to coordinate j
Ω_i	=	speed of motor i
$\boldsymbol{F}_i$	=	thrust vector generated by propeller i in the $\hat{b}_3$ direction with magnitude F_i
D_i	=	drag of propeller i

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$\boldsymbol{T}$	=	tension between the payload and the quad along the $\pm\hat{c}_3$ direction
τ_j	=	torque on the quad about the jth axis in the body frame

II. Introduction and Related Work

The topic of aircraft control with suspended loads is an area of active research interest among the controls community. Research on this subject encompasses the classic helicopter slung load problem initially studied in [1], that used linear control methods to maintain flight stability while minimizing swing during applications such as cargo transportation and disaster response scenarios. In the robotics community, the utilization of aerial robot swarms to transport loads [2], has also gained attention in the past decade. Related works on distributed control in this context often brings the advantage of increasing payload capacity and safety against failures. Nevertheless, this cable-loaded aircraft model sometimes lacks reliability and efficiency. Concerns such as payload collisions, detachment, and loading inefficiencies during takeoff, landing and transport have motivated the exploration of alternative methods.

In this context, quadcopter technologies offer viable alternatives for solving load transportation problems. With off-centered mounting of propellers, this class of UAVs makes top-mounted payload configurations a feasible option during transport. In this paper, we propose a simplified model where an inverted pendulum is mounted on the centroid of a quadcopter through a universal joint. This mechanical system was first introduced as the flying inverted pendulum in [3]. Hehn et al. approached this control problem by first linearizing the model about the upward equilibrium along some nominal trajectories. They then employed a LQR compensator to stabilize the dynamics based on the linearized system. However, general linear control approaches are vulnerable to failure if the system deviates from the linearized equilibria, for example, in a strong wind field. The problem was addressed in the literature via nonlinear control techniques, for instance, those surveyed in [4, 5], in which backstepping and feedback linearization approaches were designed to increase the basin of attraction of the inverted pendulum and to potentially achieve swing-up motion. Still, such model-based control methods heavily rely on prior information about the model and may not be suitable in cases where model uncertainties or disturbances are significant. This shortcoming becomes more evident when it comes to accounting for payload weight, which can often be challenging to determine in advance, or may change with time, such as during aerial firefighting operations.

To address the discrepancy between the theoretical model and real-world dynamics, a few recent research projects have pivoted towards data-driven methods to "learn" the system dynamics. In recent literature, Gaussian Process methods have been shown to be effective for agile and high speed maneuvers, given unknown aerodynamic effect in applications such as drone racing [6]. Data driven methods are also applicable in quadcopter payload problems when the payload weight is not known in advance, such as the HAVOK technique [7] and the CAFVI reinforcement learning scheme [8]. Still, the extensive offline training process required by those approaches may lead to slow convergence and failure in capturing time varying uncertainties. In contrast, while traditional adaptive control methods necessitate a predefined parameterized uncertainty model, the online estimation capabilities offer improved robustness against time changing uncertainties [9–11].

Among the adaptive control methods, the $\mathcal{L}_1$ adaptive controller has emerged as a particularly effective solution for UAV applications [12–18]. Its growing popularity can be attributed to its fast adaptation compared to traditional methods such as model reference adaptive control [19, 20]. The $\mathcal{L}_1$ controller also offers the advantage of performance and robustness separation by augmenting the output of a baseline controller instead of being fully integrated with it. The performance of control system is thus predominantly influenced by the baseline controller. Research on different baseline controllers detailed in [6, 21] has demonstrated that when nonlinear MPC is employed as the baseline, there is a notable reduction in tracking error during high speed flights compared to alternative controllers such as INDI and MPPI. In our work, we leverage such insights to further investigate the integration of the $\mathcal{L}_1$ adaptive control law with a nonlinear MPC to achieve maneuvering of the flying inverted pendulum while simultaneously maintaining load balance.

The remaining content of the paper is organized as follows. In section III, we introduce the dynamics model by defining the reference frames and coordinates. Subsequently, we will utilize the Euler-Lagrangian formulation to derive the equations of motion dominating the system. In section IV, our overall control architecture, as well as the MPC and $\mathcal{L}_1$ controller design principles are presented. Finally, in section V, we describe the experimental setup, validate the effectiveness of our proposed controller, and discuss comparisons with other existing nonlinear control methods.

III. Preliminaries

Deriving the dynamics model of the system is an essential first step towards synthesizing model-based control methods. To construct the mathematical model, we use Euler-Lagrangian method, with notations and the derivation process adapted from [22–24]. While formulating the full dynamics behavior in real-world environments may be complex and intractable, a few assumptions are introduced to greatly simplify the dynamical model without significantly compromising its applicability to real-world scenarios.

A. Assumptions

The following statements are assumed in later discussions of the dynamics behavior of the system.

1. The payload is a point mass.
2. Rod connecting the payload on the pendulum is rigid and mass-less.
3. Aerodynamics effect other than propeller thrust and drag are neglected.
4. The quadcopter frame is symmetric and rigid.
5. All four propellers are identical.
6. Changes in control inputs are immediately reflected in the dynamics without latency.
7. The universal joint is frictionless

Note that, unlike the assumptions made in [3], the pendulum dynamics does affect the quadcopter and is not neglected in the equations of motion.

B. Vehicle Dynamics

To model the quadcopter vehicle dynamics, two ortho-normal frames of reference other than the inertial frame are introduced. The first one being the body frame $\mathcal{B}$ fixed to the quadcopter, whose origin coincides with the center of gravity $\boldsymbol{G}$ of the quadcopter. The frame is defined with a '+' configuration such that $\boldsymbol{b}_1$ points to the motor 1, and $\boldsymbol{b}_2$ is points to motor 2. $\boldsymbol{b}_3$ is then defined based on the right hand rule. An illustration of this frame is shown in Figure 1.

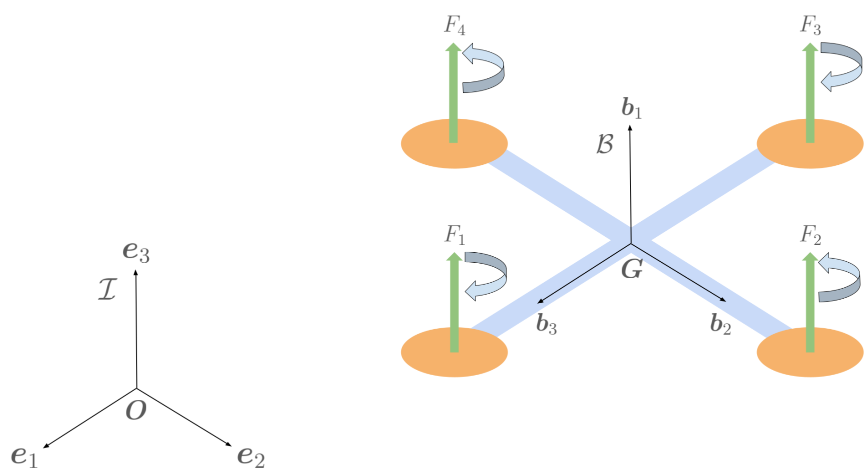


Fig. 1 Body frame $\mathcal{B}$ and free body diagram of the quadcopter (omitting weight and tension forces)

The position of $\boldsymbol{G}$ with respect to the inertial frame is expressed through the coordinates $\boldsymbol{r}_{G/O} = [x, y, z]^T$. The translational velocity is then the time derivative of position, given by ${}^{\mathcal{I}}\boldsymbol{v}_{G/O} = [\dot{x}, \dot{y}, \dot{z}]^T$, where the superscript denotes that the derivative is taken in basis $\mathcal{I}$. In the remainder of this paper, all quantities pertaining to translational velocities of the system are assumed to be with respect to the inertial frame, and the superscript will be dropped for simplicity. The orientation of the body frame with respect to the inertial frame is determined through a set of 3-2-1 Euler angles $\boldsymbol{\varphi} = [\psi, \theta, \phi]^T$. This sequence of rotation defines a rotation matrix that represents the orientation of frame $\mathcal{B}$ with respect to the inertial frame, given by

$$\boldsymbol{R}_{\mathcal{B}}^{\mathcal{I}} = \boldsymbol{R}_{3,\psi}\, \boldsymbol{R}_{2,\theta}\, \boldsymbol{R}_{1,\phi} = \begin{bmatrix} c\psi\, c\theta & c\psi\, s\phi\, s\theta - c\phi\, s\psi & s\phi\, s\psi + c\phi\, c\psi\, s\theta \\ c\theta\, s\psi & c\phi\, c\psi + s\phi\, s\psi\, s\theta & c\phi\, s\psi\, s\theta - c\psi\, s\phi \\ -s\theta & c\theta\, s\phi & c\phi\, c\theta \end{bmatrix} = [\boldsymbol{b}_1\,, \boldsymbol{b}_2\,, \boldsymbol{b}_3] \tag{1}$$

where $\boldsymbol{b}_{\cdot}$ are the columns of the rotation matrix and can be thought of as the basis of $\mathcal{B}$ represented in the inertial basis. Moreover, we choose to express the angular velocities of the quadcopter by a body rate vector, which can be tied to the definition of frame $\mathcal{B}$ and written in terms of Euler rates, given by

$${}^{\mathcal{I}}\boldsymbol{\omega}^{\mathcal{B}} = \left(\boldsymbol{R}_{2,\theta}\, \boldsymbol{R}_{1,\phi}\right)^{-1} \begin{bmatrix} 0 \\ 0 \\ \dot{\phi} \end{bmatrix} + \left(\boldsymbol{R}_{1,\phi}\right)^{-1} \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix} + \begin{bmatrix} \dot{\psi} \\ 0 \\ 0 \end{bmatrix} = \boldsymbol{W} \begin{bmatrix} \dot{\psi} \\ \dot{\theta} \\ \dot{\phi} \end{bmatrix} = \begin{bmatrix} \dot{\phi} - \dot{\psi}\, s\theta \\ \dot{\theta}\, c\phi + \dot{\psi}\, c\theta\, s\phi \\ -\dot{\theta}\, s\phi + \dot{\psi}\, c\theta\, c\phi \end{bmatrix}, \tag{2}$$

where $\boldsymbol{W}$ denote the transformation from the Euler rates to the body rates, and the superscripts indicate that the quantity represents the angular rate of the body frame with respect to the inertial frame of reference. For simplicity, the superscripts here are also dropped in later sections.

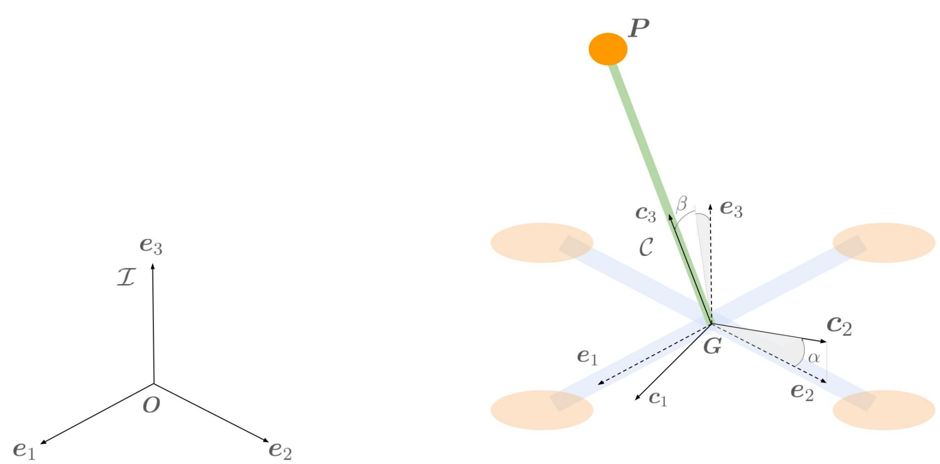


Fig. 2 Frame $\mathcal{C}$ attached to the pendulum

The second frame $\mathcal{C}$ is attached to the inverted pendulum with its origin at $\boldsymbol{G}$ (see Figure 2). The orientation of this frame is derived from an 'α-rotation' about $\boldsymbol{e}_1$, followed by a 'β-rotation' about the second axis of the intermediate frame. The resultant frame would the have $\boldsymbol{c}_3$ aligned with the pendulum, and the vector $[\alpha\,, \beta]^T$ is able to fully specify the position of the payload. The rotation matrix that represents this frame is given by

$$\boldsymbol{R}_{\mathcal{C}}^{\mathcal{I}} = \boldsymbol{R}_{1,\alpha}\, \boldsymbol{R}_{2,\beta} = \begin{bmatrix} c\beta & 0 & s\beta \\ s\alpha\, s\beta & c\alpha & -c\beta\, s\alpha \\ -c\alpha\, s\beta & s\alpha & c\alpha\, c\beta \end{bmatrix} = [\boldsymbol{c}_1,\, \boldsymbol{c}_2,\, \boldsymbol{c}_3]\,. \tag{3}$$

where $\boldsymbol{c}_{\cdot}$ are the columns of the rotation matrix. The position of the payload with respect to $\boldsymbol{G}$ is

$$\boldsymbol{r}_{P/G} = \boldsymbol{R}_{\mathcal{C}}^{\mathcal{I}} \begin{bmatrix} 0 \\ 0 \\ l \end{bmatrix} = \begin{bmatrix} l\, c\alpha\, s\beta \\ l\, s\alpha\, s\beta \\ l\, c\beta \end{bmatrix}, \tag{4}$$

where l is the length of the pendulum. Another important quantity in this system is the velocity of the payload with respect to the inertial frame, which is the vector sum of quadcopter's translational velocity and the velocity of the payload with respect to the body frame.

$$\boldsymbol{v}_{P/O} = \frac{{}^{\mathcal{I}}d}{dt}\left(\boldsymbol{r}_{P/G}\right) + \boldsymbol{v}_{G/O} = \begin{bmatrix} \dot{x} - l\,\dot{\alpha}\,s\alpha\,s\beta + l\,\dot{\beta}\,c\alpha\,c\beta \\ \dot{y} + l\,\dot{\alpha}\,c\alpha\,s\beta + l\,\dot{\beta}\,c\beta\,s\alpha \\ \dot{z} - l\,\dot{\beta}\,s\beta \end{bmatrix} \tag{5}$$

Together, the system configuration can be specified through 8 generalized coordinates $\boldsymbol{q} = [x\,, y\,, z\,, \psi\,, \theta\,, \phi\,, \alpha\,, \beta]^T$. From here, we can compute the Lagrangian as the difference between the kinetic energy and the potential energy of the system, given by

$$\mathcal{L} = \frac{1}{2}\left[m_G\left\|\boldsymbol{v}_{G/O}\right\|^2 + \boldsymbol{\omega}^T\boldsymbol{I}_G\,\boldsymbol{\omega}\right] + \frac{1}{2}m_P\left\|\boldsymbol{v}_{P/O}\right\|^2 - m_G\left(\boldsymbol{g}\cdot\boldsymbol{r}_{G/O}\right) - m_P\left(\boldsymbol{g}\cdot\boldsymbol{r}_{P/O}\right), \tag{6}$$

where $\boldsymbol{I}_G$ denotes the principal moment of inertia matrix, m_G, m_P denote the mass of the quadcopter and pendulum, and $\boldsymbol{g} = [0\ 0\ g]^T$ is the gravity vector.

The left hand side of the Euler-Lagrangian Equation is then defined based on Eq. 6, and the right hand side of the equations are determined by the generalized forces of this system. Note that due to symmetry, the translation dynamics of the quadcopter are independent of the angular rates, and the assumption of a friction-less joint implies that the pendulum dynamics are decoupled from the rotational dynamics. Therefore, a simpler way to describe the rotational equations of motion is discussed later in the section and omitted here. The coordinates of interest are now reduced to the subset $\mathbf{q} = [x,\ y,\ z,\ \alpha,\ \beta]^T$, and based on assumptions in section III.A, the only non-conservative forces in the system include the tension force along the pendulum, the thrust forces $F_1,\ F_2,\ F_3,\ F_4$, which all point towards the $\boldsymbol{b}_3$ direction, as well as the drag forces from each propeller. Therefore, the generalized forces $\boldsymbol{Q}$ for the remaining coordinates are defined using D'Almbert's Principle, given by

$$\boldsymbol{Q} = \begin{bmatrix} \boldsymbol{R}_{\mathcal{B}}^{\mathcal{I}}\boldsymbol{T} \\ \boldsymbol{0}_{2\times 1} \end{bmatrix} \tag{7}$$

where $\boldsymbol{T} = [0,\ 0,\ T]^T$ denotes the forces on the quadcopter in the three body frame directions, and T denotes the total thrust produced by the 4 rotors. Together, the nonlinear dynamic equations for can be derived using the Euler-Lagrange Equation, stated as

$$\frac{d}{dt}\frac{\partial\mathcal{L}}{\partial\dot{\boldsymbol{q}}} - \frac{\partial\mathcal{L}}{\partial\boldsymbol{q}} = \boldsymbol{Q}. \tag{8}$$

Upon algebraic manipulations to separate some of the second order terms, Eq. 8 can be expanded to

$$\ddot{x} = \frac{l\,m_P}{m_G + m_P}\left(\dot{\beta}^2 s\beta - \ddot{\beta}c\beta\right) + \frac{T}{m_G + m_P}\left(s\psi s\phi + c\psi s\theta c\phi\right) \tag{9}$$

$$\ddot{y} = \frac{l\,m_P}{m_G + m_P}\left(\dot{\alpha}^2 s\alpha c\beta + \dot{\beta}^2 s\alpha c\beta - \ddot{\alpha}c\alpha c\beta + \ddot{\beta}s\alpha s\beta + 2\dot{\alpha}\dot{\beta}c\alpha s\beta\right) + \frac{T}{m_G + m_P}\left(s\psi s\theta c\phi - c\psi s\phi\right) \tag{10}$$

$$\ddot{z} = \frac{l\,m_P}{m_G + m_P}\left(\dot{\alpha}^2 c\alpha c\beta + \dot{\beta}^2 c\alpha c\beta + \ddot{\alpha}c\beta s\alpha + \ddot{\beta}c\alpha s\beta - 2\dot{\alpha}\dot{\beta}s\alpha s\beta\right) + \frac{T}{m_G + m_P}\left(c\theta c\phi\right) - g \tag{11}$$

$$\ddot{\alpha} = \frac{\ddot{y}\,c\alpha + \ddot{z}\,s\alpha + 2l\,\dot{\alpha}\,\dot{\beta}\,s\beta + g\,s\alpha}{l\,c\beta} \tag{12}$$

$$\ddot{\beta} = \dot{\alpha}^2\,c\beta\,s\beta - \frac{\ddot{x}\,c\beta + \ddot{y}\,s\alpha\,s\beta - \ddot{z}\,c\alpha\,s\beta - g\,c\alpha\,s\beta}{l} \tag{13}$$

From assumption 7., it is obvious that the rotational dynamics of the quadcopter is decoupled from the motion of the pendulum. Thus, employing the Newton-Euler Equation presents a more concise way to describe rotations of the quad in terms of body angular rates instead of Euler angles. From here, the full state transition equations can be obtained by further separating the second order terms, which is simplified as

$$\begin{bmatrix} \dot{\boldsymbol{r}} \\ \dot{\boldsymbol{\varphi}} \\ \dot{\alpha} \\ \dot{\beta} \\ \dot{\boldsymbol{v}} \\ \dot{\boldsymbol{\omega}} \\ \ddot{\alpha} \\ \ddot{\beta} \end{bmatrix} = \begin{bmatrix} \boldsymbol{v} \\ \boldsymbol{W}^{-1}\boldsymbol{\omega} \\ \dot{\alpha} \\ \dot{\beta} \\ \left[\boldsymbol{R}_{\mathcal{B}}^{\mathcal{I}}\boldsymbol{T} + l m_P \boldsymbol{c}_3(\dot{\alpha}^2 c\beta^2 + \dot{\beta}^2)\right]/(m_G + m_P) - \boldsymbol{g} \\ \boldsymbol{I}^{-1}\left[\tau - \boldsymbol{\omega} \times \boldsymbol{I}\boldsymbol{\omega}\right] \\ \boldsymbol{c}_2^T \boldsymbol{R}_{\mathcal{B}}^{\mathcal{I}}\boldsymbol{T}/[l(m_G + m_P)c\beta] + 2\dot{\alpha}\dot{\beta}\, s\beta/c\beta \\ -\boldsymbol{c}_1^T \boldsymbol{R}_{\mathcal{B}}^{\mathcal{I}}\boldsymbol{T}/[l(m_G + m_P)] - \dot{\alpha}^2 s(2\beta)/2 \end{bmatrix}, \tag{14}$$

where $\boldsymbol{\tau}$ is the torque induced by the rotors about the three body frame axes and $\boldsymbol{v}$ is the abbreviation for $\boldsymbol{v}_{G/O}$.

Moreover, with the direction of motor spin and thrust vectors illustrated in figure 1, a control allocation matrix is used to transform the motor inputs to the forces that matches the form present in the equations of motion. Specifically, with the assumptions that all rotors are identical and that thrust and drag are proportional to the motors speed squared, the relationship is given by

$$\begin{bmatrix} T \\ \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix} = \begin{bmatrix} C_L & C_L & C_L & C_L \\ 0 & d\,C_L & 0 & -d\,C_L \\ -d\,C_L & 0 & d\,C_L & 0 \\ C_D & -C_D & C_D & -C_D \end{bmatrix} \begin{bmatrix} \Omega_1^2 \\ \Omega_2^2 \\ \Omega_3^2 \\ \Omega_4^2 \end{bmatrix}, \tag{15}$$

where d denotes the arm length of the quadcopter (half of the wheelbase), Ω denotes each of the 4 motor speeds, and C_L, C_D denote the lift and drag coefficients of the propellers, respectively.

IV. Methodology

This section presents an adaptive model-predictive control (AMPC) algorithm to stabilize a top-loaded UAV along specified trajectories given model uncertainties and external disturbances. The control structure consists of two distinct components. The baseline is a nonlinear model-predictive controller similar to that in [21, 25, 26]. The MPC solves the optimal control problem (OCP) recursively given an additive cost function and the dynamics model given by Equations 14 to ensure effective reference tracking of the system.

Upon generation of the optimal trajectory, the MPC output is then passed into the $\mathcal{L}_1$ controller that tracks a nonlinear reference model [12]. Specifically, with the state feedback and MPC control commands as inputs for the reference model, the $\mathcal{L}_1$ controller then uses a piecewise constant adaptive law to perform online estimation of both matched and unmatched uncertainties in the uncertainty model [27–29].The uncertainties are then passed into a low pass filter and augmented with the MPC output via addition. Similar derivation of the adaptive controller has been studied in [13, 14, 21].

In our implementation, we integrate the controller design principles from existing methods to formulate a controller that is capable of both stabilization and trajectory-tracking tasks. A block diagram of this is shown in Figure 3.

A. Model Predictive Control

The model predictive controller combines open loop trajectory optimization with feedback control. At each control step, given the state feedback, the MPC algorithm solves an OCP with additive cost function J to obtain a series of piecewise constant optimal inputs $\boldsymbol{u}_{\text{opt}}$ over a finite prediction horizon N. Once optimal control action is solved, depending on the control horizon N_c, the controller would actuate the first N_c steps of the computed inputs, then repeat the same procedure at the next control time step.

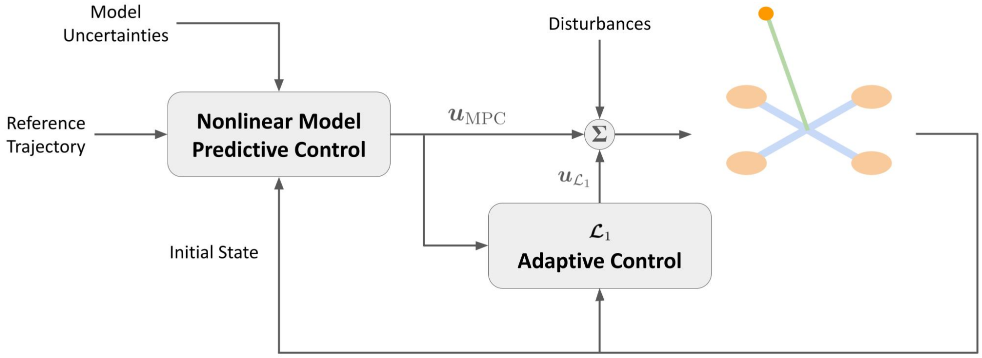


Fig. 3 Block diagram of the adaptive model-predictive control law.

A crucial constraint imposed on the OCP is the dynamics of the system. Specifically, with the control actions as part of the decision variable, the optimization solver numerically computes the predicted system trajectory using the modelled state transition equations and set such prediction as a constraint on the optimization problem to ensure that the solution is physically feasible. While the system dynamics is continuous in nature, setting the inputs and dynamic constraints as continuous time, infinite dimension functions makes the optimization problem intractable. One popular class of methods to reduce the problem scale is the direct shooting technique studied in [30], which discretizes the decision variables by forward simulating the system dynamics through a Runge-Kutta solver to obtain the predicted state trajectory $z(t_k)$, $k \in [1, \cdots, N]$ at discrete time steps provided the initial state z_0. Still, in large-scale problems with more than a few number of states, this method becomes inefficient and the repeated call of the integrator adds to computation time. Therefore, in our implementation, the direct collocation method is employed, which uses N discrete state and input vectors as decision variables that is then interpolated, with cubic splines and first order polynomials, to obtain a smooth time continuous trajectory [31]. Subsequently, equality constraints are applied at the collocation points $t_{c,k}$, where the time derivative of the cubic spline must match the dynamics stated in Equations 14. Choosing the collocation point as the middle between two time steps $t_{c,k} = (t_k + t_{k+1})/2$, these constraints are given by

$$\boldsymbol{u}(t_{c,k}) = \frac{1}{2}\left(\boldsymbol{u}(t_k) + \boldsymbol{u}(t_{k+1})\right) \tag{16}$$

$$z(t_{c,k}) = \frac{1}{2}\left(z(t_k) + z(t_{k+1})\right) + \frac{h_k}{8}\left(\dot{z}(t_k) - \dot{z}(t_{k+1})\right) \tag{17}$$

$$\dot{z}(t_{c,k}) = -\frac{3}{2h_k}\left(z(t_k) - z(t_{k+1})\right) - \frac{1}{4}\left(\dot{z}(t_k) + \dot{z}(t_{k+1})\right). \tag{18}$$

where $h_k = t_{k+1} - t_k$ is the step size at step k. Note that since this constraint is simply an approximation, it only works in small time intervals and the prediction time steps needs to be sufficiently small. With the discretized decision variables, the additive cost function is then defined with a quadratic form, given by

$$J = \boldsymbol{x}_N^T \boldsymbol{F} \boldsymbol{x}_N + \sum_{k=0}^{N_p-1} \boldsymbol{x}_k^T \boldsymbol{Q} \boldsymbol{x}_k + \boldsymbol{u}_k^T \boldsymbol{R} \boldsymbol{u}_k \tag{19}$$

$$\boldsymbol{x}_k = z(t_k) - z_d(t_k) \tag{20}$$

$$\boldsymbol{u}_k = \boldsymbol{\mu}(t_k) - \boldsymbol{\mu}_d(t_k) \tag{21}$$

where $\boldsymbol{x}$ is the tracking error, $\boldsymbol{F}$ is the terminal cost matrix, $\boldsymbol{Q}$ and $\boldsymbol{R}$ are semi-positive definite error and actuation cost matrices, respectively.

Together, the OCP is formally stated as

$$
\begin{aligned}
\min_{z,\boldsymbol{u}} \quad & \boldsymbol{x}_N^T \boldsymbol{F} \boldsymbol{x}_N + \sum_{k=0}^{N_p-1} \boldsymbol{x}_k^T \boldsymbol{Q} \boldsymbol{x}_k + \boldsymbol{u}_k^T \boldsymbol{R} \boldsymbol{u}_k \\
s.t. \quad & \dot{z}(t_k) = f(z(t_k), \boldsymbol{u}(t_k)), \quad \forall i \in [0, N-1] \\
& z(0) = z_0 \\
& \mathbf{u} \in [\mathbf{u}_{\min},\ \mathbf{u}_{\max}] \\
& \mathbf{z} \in [\mathbf{z}_{\min},\ \mathbf{z}_{\max}],
\end{aligned}
\tag{22}
$$

where $\mathbf{u}_{\min},\ \mathbf{u}_{\max}$ are the input saturation, $\mathbf{z}_{\min},\ \mathbf{z}_{\max}$ are the state bounds, and z_0 is the initial condition.

Given the nonlinearities in the dynamics, we choose to implement sequential quadratic programming (SQP) algorithm to solve the OCP. This method, which appends the cost function with the constraints to form a Lagrangian, then approximates the Lagrangian with a second order Taylor expansion. Explanation of the algorithm is detailed in [32]. While SQP and most optimization algorithm requires an initial guess of the decision variables, in our implementation, it is set in a recursive manner. The initial guess at the start of the program is set as the state and input trajectories observed when using the LQR compensator. For any future iterations of the MPC, since the system is only shifted N_c time steps, we assume that the current optimal solution is in the proximity of the previous one, and thus the initial guess is set to the result of the previous optimal solution to speed up the optimization and reduce required exploration in the objective space.

B. $\mathcal{L}_1$ Adaptive Control

The $\mathcal{L}_1$ adaptive controller is appended to the MPC output to account for changing dynamics and uncertainties in the model. Since there is no uncertainties in first order terms the interested state variables are reduced to $z = [\boldsymbol{v}, \boldsymbol{\omega}]$. The pendulum dynamics are also dropped from the state variables considering the main objective of position reference tracking. Therefore, the nonlinear reference model with desired dynamics and no uncertainty is derived from Equations 14 as

$$
\boldsymbol{f}(t) = \begin{bmatrix} \dot{\boldsymbol{v}} \\ \dot{\omega} \end{bmatrix} = \begin{bmatrix} \frac{\boldsymbol{R}_{\mathcal{B}}^{\mathcal{I}} \boldsymbol{T}_{\text{MPC}}}{m_P + m_G} + \frac{l m_P \boldsymbol{c}_3 (\dot{\alpha}^2 c\beta^2 + \dot{\beta}^2)}{m_P + m_G} - \boldsymbol{g} \\ \boldsymbol{I}^{-1} \left[\boldsymbol{\tau}_{\text{MPC}} - \boldsymbol{\omega} \times \boldsymbol{I} \boldsymbol{\omega} \right] \end{bmatrix}. \tag{23}
$$

Note that, as stated in Section III.A, air resistance is neglected in the reference model considering the small form factor of the pendulum and the low speed applications which the vehicle is mostly likely deployed in. Thus, all aerodynamic effects are considered uncertainties that are compensated by the $\mathcal{L}_1$ control output. Let $\boldsymbol{\varsigma} = [\varsigma_x, \varsigma_y, \varsigma_z]^T$ be uncertainties in the quadcopter's linear acceleration which also encompasses uncertainties in the pendulum's mass and length, and $\boldsymbol{\xi} = [\xi_x, \xi_y, \xi_z]^T$ be uncertainties in the quadcopter's angular acceleration. The uncertainty model can then be written as

$$
\dot{\boldsymbol{v}} = \frac{\boldsymbol{R}_{\mathcal{B}}^{\mathcal{I}}}{m_P + m_G} \left[\boldsymbol{T} + \boldsymbol{\varsigma} \right] + \frac{l m_P \boldsymbol{c}_3 (\dot{\alpha}^2 c\beta^2 + \dot{\beta}^2)}{m_P + m_G} - \boldsymbol{g} \tag{24}
$$

$$
\dot{\omega} = \boldsymbol{I}^{-1} \left[\boldsymbol{\tau} - \omega \times \boldsymbol{I} \omega + \boldsymbol{\xi} \right] \tag{25}
$$

Since the quadcopter is underactuated and the propeller thrusts can only be provided in the z axis direction, uncertainties in x and y linear accelerations are not directly controllable and are thus considered unmatched uncertainties, defined as $\boldsymbol{\sigma}_{um} \triangleq [\varsigma_x, \varsigma_y]^T$. The remaining uncertainties can directly be compensated and are matched uncertainties $\boldsymbol{\sigma}_m \triangleq [\varsigma_z, \xi_x, \xi_y, \xi_z]^T$ uncertainties. From here, Equations 24 can be further broken down into the reference, matched, and unmatched uncertainties, given by

$$
\dot{z} = \boldsymbol{f}(t) + \boldsymbol{g}(t)(\boldsymbol{u}_{\mathcal{L}1} + \boldsymbol{\sigma}_m) + \boldsymbol{g}^{\perp}(t) \boldsymbol{\sigma}_{um} \tag{26}
$$

where $\boldsymbol{g}, \boldsymbol{g}^{\perp}$ are perpendicular matrices given by

$$g(t) = \begin{bmatrix} \frac{\boldsymbol{e}_z^{\mathcal{B}}}{m_P+m_G} & \mathbf{0}_{3\times 3} \\ \mathbf{0}_{3\times 1} & \boldsymbol{I}^{-1} \end{bmatrix} \tag{27}$$

$$g^{\perp}(t) = \begin{bmatrix} \frac{\boldsymbol{e}_x^{\mathcal{B}}}{m_P+m_G} & \frac{\boldsymbol{e}_y^{\mathcal{B}}}{m_P+m_G} \\ \mathbf{0}_{3\times 1} & \mathbf{0}_{3\times 1} \end{bmatrix} \tag{28}$$

To estimate the uncertainties, we define the $\mathcal{L}_1$ state observer as

$$\dot{\hat{\boldsymbol{z}}} = \boldsymbol{f}(t) + \boldsymbol{g}(t)(\boldsymbol{u}_{\mathcal{L}_1} + \hat{\boldsymbol{\sigma}}_m) + \boldsymbol{g}^{\perp}(t)\hat{\boldsymbol{\sigma}}_{um} + \boldsymbol{A}_s(\boldsymbol{z} - \hat{\boldsymbol{z}}) \tag{29}$$

where $\hat{\boldsymbol{\sigma}}_m, \hat{\boldsymbol{\sigma}}_u m$ are the estimated uncertainties, $\hat{\boldsymbol{z}}$ is the predicted states, and $\boldsymbol{A}_s$ is a Hurwitz matrix used to tune the controller. The adaptive law is then defined by

$$\begin{bmatrix} \hat{\boldsymbol{\sigma}}_m(iT_s) \\ \hat{\boldsymbol{\sigma}}_{um}(iT_s) \end{bmatrix} = -\mathbb{I}\boldsymbol{G}^{-1}(iT_s)\boldsymbol{\Phi}^{-1}\mu(iT_s), \tag{30}$$

where $\boldsymbol{G}(iT_s) = [\boldsymbol{g}(iT_s), \boldsymbol{g}^{\perp}(iT_s)]^T$ and $\mu(iT_s) = e^{\boldsymbol{A}_s T_s}[\boldsymbol{z}(iT_s) - \hat{\boldsymbol{z}}(iT_s)]$ are evaluated at the ith time step and $\boldsymbol{\Phi} = \boldsymbol{A}_s^{-1}\left(e^{\boldsymbol{A}_s T_s} - \mathbb{I}\right)$. From here, the matched and unmatched uncertainties can be compensated through the control law given by

$$\boldsymbol{u}_{\mathcal{L}_1}(s) = -\boldsymbol{C}(s)\left[\hat{\boldsymbol{\sigma}}_m + H_m^{-1}(s)H_{um}(s)\hat{\boldsymbol{\sigma}}_{um}\right], \tag{31}$$

where $C(s)$ is a strictly proper first order low pass filter, and H_m, H_{um} are transfer matrices defined by

$$H_m(s) = [\mathbf{0}_{4\times 2} \quad \mathbb{I}_4]\,(s\mathbb{I}_6 - A_s)^{-1}g \tag{32}$$

$$H_{um}(s) = [\mathbb{I}_2 \quad \mathbf{0}_{4\times 4}]\,(s\mathbb{I}_6 - A_s)^{-1}g^{\perp}. \tag{33}$$

The control law can then be converted to discrete time through Z-transforms, which yields a recursive formula programmable in practice.

V. Experiments and Results

A. Simulation

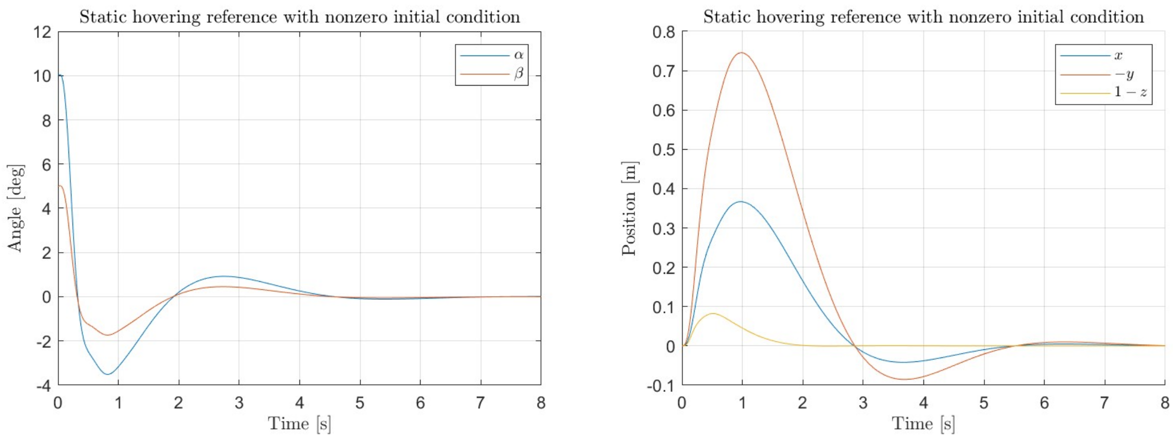
The simulation environment and visualizations are developed in MATLAB with the state trajectory obtained numerically through the built-in Runge-Kutta 45 solver. The physical parameters of the quadcopter, as listed in Table 1, are from the custom drone developed by the Motion and Teaming Lab in the University of Maryland, College Park.

Quantity	Value	Unit
m_G	0.9	kg
d	0.15	m
I_x, I_y	0.0045	kg m^2
I_z	0.01	kg m^2
C_D	4.4E-9	N s^2/rad^2
C_L	2.7E-8	Nm s^2/rad^2
l	1.5	m
m_P	0.05	kg

Table 1 Physical Parameters of the vehicle.

To verify the functionality of our controller under uncertainties, two test scenarios are employed. Firstly, an ideal model that matches the real-world is used, and the system response from a nonzero initial condition is shown in Figure 4.

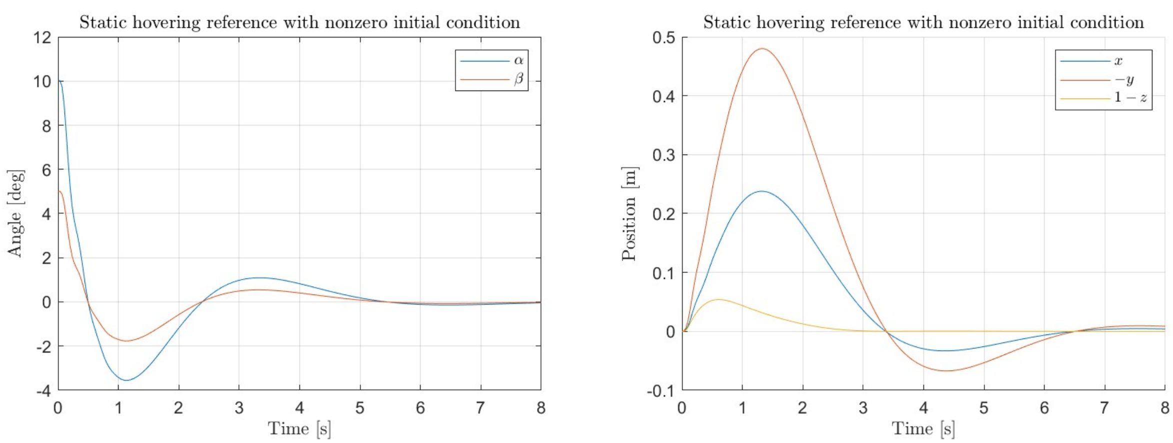
We can observe that the system reaches 99% of the steady state around 6 seconds and the maximum displacement is around 0.75 m, which well fits the size of the testing arena if a physical experiment were to be carried out.



(a) Pendulum angle **(b) Position**

Fig. 4 System Response with static hovering reference at altitude of 1 m. The initial pendulum angles are 5° **and** 10° **for** α **and** β **respectively.**

The second test case is when the pendulum configuration deviates from the model. The mass used in the simulation is now 0.2 kg and the length is 1 m, while the values in Table 1 are still used by the MPC solver. The system response in this case is shown in Figure 5. We can observe that the system experiences a slight larger rise time while the adaptive controller is still learning the uncertainties. As the adaptive law converges, the system's behavior becomes more align with the ideal model and the settling time is only increased by around 1 second.



(a) Pendulum angle **(b) Position**

Fig. 5 System Response with static hovering reference at altitude of 1 m given model uncertainty. The initial pendulum angles are also 5° **and** 10° **for** α **and** β **respectively.**

B. Performance Comparisons

A major question this paper tries to resolve is how well the proposed controller perform compared to other existing control methods. Four different control approaches are considered. The first two control methods are $\mathcal{L}_1$ augmented AMPC and non-adaptive MPC, with the control parameters carried over from that used in Section V.A. The third controller is the linear quadratic regulator studied in [3]. Specifically, the system dynamics is linearized about hover with the pendulum at the upright equilibrium, and and the new equations of motion are given by

$$\begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \\ \dot{\boldsymbol{\omega}} \\ \ddot{\alpha} \\ \ddot{\beta} \end{bmatrix} = \begin{bmatrix} g\left(-m_P\beta - \theta\right)/m_G \\ g\left(m_P\alpha - \psi\right)/m_G \\ \left(T - g\right)/\left(m_G + m_P\right) \\ \boldsymbol{I}^{-1}\boldsymbol{\tau} \\ g\left[(m_G + m_P)\alpha - \psi\right]/\left(l m_G\right) \\ g\left[(m_G + m_P)\beta - \theta\right]/\left(l m_G\right) \end{bmatrix}, \tag{34}$$

which can then be written into standard form $\dot{\boldsymbol{x}} = A\boldsymbol{x} + B\boldsymbol{u}$ and used in the Riccati equation to solved for the infinite horizon LQR gains.

The fourth controller utilizes the feedback linearization method detailed in [5, 33]. The control law is synthesized with 3 control loops, with the most output loop tracking the pendulum angles, the middle loop tracking position, and the inner loop tracking the quadcopter orientation.

Three different metrics are consider to measure the performance of the control methods with parameters in Table 1 and uncertatinties described earlier used. The first basis is the settling time given a step input from position [0,0] to [1,1]. The settling time is measured as the time after which both x and y reached 99% of steady state, and performances of the tested controllers are listed in Table 2.

Controller	**AMPC**	**MPC**	**LQR**	**NDI**
t_s [s]	6.9	12.3	9.0	9.8

Table 2 Settling time of the tested controllers in seconds: time taken by the controller to drive ths system within 99% of desired reference state.

Such result shows that the proposed controller is robust against uncertainties in the model and enjoys a much faster settling time. In contrast, the other are highly model-based control methods and they suffer from deviant behaviors from the ideal model. This is especially true for MPC whose required computation limits the control bandwidth and thus limiting the control authority to rectify errors from the reference trajectory.

The second metric is the basin of attraction given various initial pendulum angles. Specifically, considering the vehicle symmetry, such the basin of attraction here is defined by the maximum initial α angle such that the close loop system is convergent and that the state trajectory is physical feasible, for instance, not the pendulum not overlapping with the quadcopter. The test output is shown is Table 3. The limited basin of attraction of LQR compensators can be attributed to its linearized reference model. Moreover, while the NDI controller is capable of dealing with nonlinearities as the pendulum deviates away from the upright, uncertainties in the model has resulted either in unrealistic motion, in this case, the quadcopter colliding with the pendulum, or complex valued inputs during the model inversion. Similarly, although MPC is a nonlinear controller, the model mismatched had lead to the optimization solver converging to infeasible points and eventually instability of the system. Adaptive MPC, on the other hand, retains the ability to correct large errors in the pendulum angle. The ability to add hard constraints on the optimization problem also resolves the problem of unrealistic motion that occurs with NDI controllers.

Controller	**AMPC**	**MPC**	**LQR**	**NDI**
$\alpha_{0,\max}$ [deg]	50°	30°	25°	35°

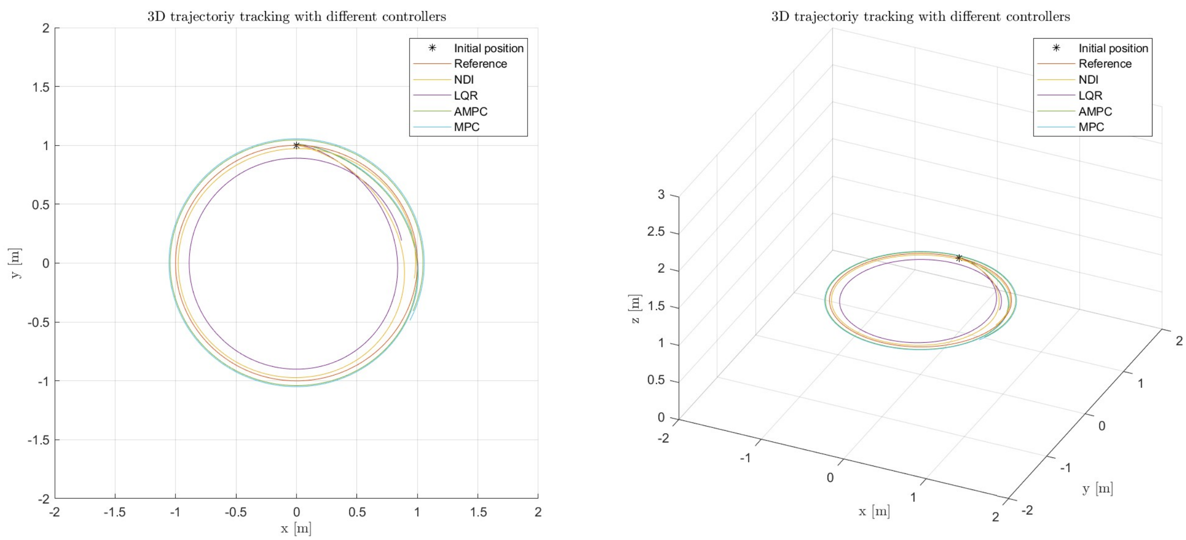
Table 3 Comparing *regions of attraction* of the tested controllers: *maximum* initial pendulum angle (off the vertical) from which the controller is able to stabilize the pendulum to upright position

The final metric is the quadcopter positional and pendulum angular tracking error given circular reference trajectories, which features a radius of 1 m and period of 8 s. The quadcopter positional and pendulum trajectories are visualized in

6, and the root-mean-square tracking error are listed in Table 4. It can be observed that the AMPC law initially follows a similar trajectory as its MPC baseline as the adaptive controller is still learning the uncertainties. At around 5 seconds, the $\mathcal{L}_1$ controller had nearly adapted to the uncertainties and was able to track the reference nearly identical as the ideal model while providing improved pendulum stabilization.

Controller	**AMPC**	**MPC**	**LQR**	**NDI**
Position RSME [m]	0.253	0.311	0.678	0.461
Pendulum RSME [deg]	1.821°	4.101°	3.389°	4.435°

Table 4 Tracking errors of the quadrotor vehicle given a circular reference.



(a) Top down view **(b) Perspective view**

Fig. 6 Quadcopter's trajectory given a circular reference.

VI. Conclusions and Future Work

In this work, we propose a novel control strategy for the flying inverted pendulum by combining direct-collocated NMPC with an $\mathcal{L}_1$ adaptive controller to simultaneously stabilize the pendulum and track a positional reference in the presence of model uncertainties. The derived adaptation law and the NMPC are both grounded in the same set of nonlinear dynamics equations that serve as the guiding framework for our system's behavior. Through simulation studies, we have demonstrated that the proposed $\mathcal{L}_1$-NMPC approach outperforms existing methods, particularly in scenarios with uncertain model parameters. In the future, our goal is to transition from simulation to real-world testing and to further explore the capabilities of our control structure in actual operational environments. With the control law validated on real systems, we can then unlock the full potential of our $\mathcal{L}_1$-NMPC system for practical applications.

VII. Appendix

The code used in this paper along with additional results and animations can be found in the following github page — `https://github.com/yhan0117/fly_inv_pend`.

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